Direct Inversion of Micellar Relaxation Kinetics from Large-Amplitude Oscillatory Dilatational Surface Rheology: Theory, Bandwidth Limits, and Robust Parameter Identification for Model Nonionic Surfactants

Short Title: Micellar Relaxation Kinetics from LAOD

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Scope

This paper is restricted to the nonionic surfactant $C_{12}E_8$ on a profile-analysis tensiometer (PAT-1M, 0.005–1.0 Hz) with a Langmuir surface equation of state (a = 0). Every quantitative claim in §3–§7 is computed for that system. The ionic surfactants (CTAB, SDS) appear only in §6, as a predictive instrument-selection criterion derived from their relaxation timescales — not as experimental validations. Multicomponent and Frumkin-interaction effects are deferred to Paper 2.

Principal limitation, stated up front: this is a methods paper validated against its own forward model. It contains no measured $C_{12}E_8$ dataset, and the τ_2 = 4 s benchmark is a literature value rather than one inverted from data. See §7.7.

Abstract

The kinetics of micellar exchange at a surfactant-laden interface are carried by the slow Aniansson–Wall relaxation time τ_2, a parameter that governs the response of foams and emulsions to any non-steady area change. It is nevertheless almost invisible to conventional dilatational rheology: in the linear regime the micellar reservoir competes with diffusion, and the two contributions cannot be separated from a single frequency response. Here we show that the phase lag of the surface-tension response in large-amplitude oscillatory dilatational rheology carries τ_2 quantitatively, and we establish the conditions under which that information can be recovered.

Using a mass-conserving reaction–diffusion model in which the micellar reservoir is an explicit species with linearised Aniansson–Wall exchange, we identify a categorical signature that requires no model: no diffusion-only monolayer can exceed a loss angle of 45° at any frequency, whereas a micellar solution routinely reaches 68–80°. Because the reservoir acts as a

lagging source term that delivers surfactant late in the expansion cycle, it adds dissipation rather than elasticity — the opposite of the intuitive expectation — and this sign is diagnostic.

We then separate detectability from invertibility and show they peak at different concentrations. The second harmonic I_2 is maximal at the CMC, where the micellar reservoir is empty and its sensitivity to τ_2 is exactly zero. The operative observable is instead the loss angle, whose sensitivity exceeds its measurement uncertainty by two orders of magnitude where the harmonic ratio's does not: across a doubling of τ_2 the ratio moves 0.007–0.019 against a noise floor of 0.025–0.05. For the model nonionic surfactant $C_{12}E_8$ on a profile-analysis tensiometer we measure a concentration inversion window of 1.2–3.0 × CMC with an interior optimum at 1.5 × CMC, and show this window is invariant across a fourfold range in the assumed limiting elasticity — it is a property of the transport problem, not of the isotherm parameterisation. A sampling density of ≥ 5 frames per cycle is required, below which the phase estimate is biased rather than noisy (164–266° against a true 68° at one frame per cycle). At the operating point τ_2 is recovered with a statistical precision of 0.2–0.3% for a known aggregation number.

Because the method requires the drive frequency to straddle the relaxation rate ($\omega\tau_2 \gtrsim 1$), it converts directly into an instrument-selection criterion: the 0.005–1 Hz band of a drop-shape tensiometer covers $C_{12}E_8$ (τ_2 = 4 s) but cannot reach CTAB (65 ms) or SDS (38 ms), which require a capillary-pressure tensiometer above ≈ 16 Hz. This explains, without invoking chemical difficulty, why low-frequency drop-shape analysis has not reported micellar exchange in fast ionic surfactants.

The precision figure is statistical. Accuracy is limited by systematic error in the aggregation number, which propagates one-for-one: a 10% error in $\bar{n}$ produces a 10% bias in τ_2,

forty-four times the statistical uncertainty. Both numbers must be quoted together, and we do so throughout.

1. Introduction

1.1 The measurement, and the parameter buried inside it

When the area of a surfactant-laden interface is driven sinusoidally, the surface tension responds at the same frequency but out of phase. For a surfactant that is only transported by diffusion, that phase lag is a nuisance: it is set by the frequency and by a single diffusion time, and the standard Lucassen–van den Tempel (LVDT) modulus accounts for it completely [2]. The practical consequence is that decades of dilatational rheology have been built on a one-parameter-per-frequency description, fitted to a diffusion coefficient, and validated by its ability to collapse data across frequency.

Above the critical micelle concentration (CMC) this description is incomplete in a way that matters. Micelles constitute a reservoir of surfactant that can dissociate and feed the interface within a single oscillation cycle. The kinetics of that exchange are characterised by the slow relaxation time τ_2 of the Aniansson–Wall stepwise association scheme [3]. τ_2 is not a nuisance parameter: it controls how quickly a micellar solution can heal a depletion layer, it governs foam and emulsion stability under dynamic area change, and it is the parameter that distinguishes a "fast" from a "slow" surfactant in any application where the interfacial area is not constant.

The difficulty is that τ_2 is essentially invisible to linear methods. In the linear regime the micellar reservoir enters as a second relaxation process with a well-defined signature, but it does so in competition with diffusion, and the two contributions are not separately identifiable from a single frequency response. The classical resolution is to measure over a wide frequency range

and look for the resolvable separation of timescales — which requires an instrument spanning the crossover, and which in practice has meant that τ_2 is reported for a handful of well-studied systems and assumed for the rest.

1.2 Why large amplitude, and why nobody has used it for this

The second harmonic of the surface-tension response is generated only when the interface is driven far enough from equilibrium that the response is genuinely nonlinear. Because it is a small signal sitting on a large one, it has usually been treated as a quality-of-measurement indicator — a warning that the strain amplitude was too high for linear analysis — rather than as a source of information.

That framing has recently begun to change. Czakaj et al. [1] measured the second harmonic ratio of the surfactant LAE across concentrations spanning the CMC and found a pronounced, non-monotonic structure: the ratio rises sharply as the CMC is approached, peaks essentially at the CMC, and declines above it. They interpret this as evidence of interfacial compositional reorganisation rather than of micellar kinetics. Whatever the correct interpretation, the result establishes the central point: in the large-amplitude regime the second harmonic carries concentration-dependent information that the fundamental does not.

What has not been established is whether that information can be inverted — whether a large-amplitude oscillatory dilatational (LAOD) measurement, of the kind a standard tensiometer can actually perform in a reasonable time, contains enough information to recover τ_2 quantitatively, and under what conditions. That is the question this paper answers.

1.3 The distinction this paper turns on: detectable is not invertible

The natural first question is whether the nonlinearity is measurable. It is easy to answer, and answering it alone leads to a systematic error of reasoning that we found ourselves making

and which, we suspect, is common: a signal that clears the noise floor is not thereby a signal that determines a parameter.

We therefore separate two criteria and insist on both:

• Detectability. The second-harmonic amplitude I_2 exceeds the instrumental noise floor. This is necessary but weak — it says the nonlinearity exists, not that it encodes τ_2.

• Invertibility. The observable changes with τ_2 by an amount large compared with its uncertainty. A detectable nonlinearity whose sensitivity to τ_2 falls below its own uncertainty is useless for inversion, and the harmonic ratio is very nearly that: across a doubling of τ_2 at fixed concentration and frequency, I_2/I_1 moves by 0.007–0.019 against a noise floor of 0.025–0.05 on the ratio itself. The signal is real but not resolvable — a weaker and more defensible claim than calling it blind.

The observable that does carry τ_2 is not the amplitude ratio but the phase lag. This is counter-intuitive — phase is usually regarded as the noisier measurement — and it is the pivot on which the whole method rests.

1.4 Why an operational window exists at all

Above the CMC the micellar reservoir grows with total concentration, so one expects the reservoir effect to become more pronounced as concentration is raised. The opposite happens. The reservoir capacity C_m enters the problem twice: it strengthens the buffering of the interfacial depletion layer, and it smears the phase response toward the diffusion-limited limit by holding the subsurface concentration near its equilibrium value. Beyond a few times the CMC the second harmonic falls below the noise floor and the phase becomes τ_2-insensitive again. The method therefore lives in a bounded window above the CMC, and — importantly for anyone who tries to reproduce it — the window has two edges, not one.

At exactly the CMC there is no micellar reservoir at all, and the phase lag becomes strictly independent of τ_2. The informative region is not "near the CMC" in the loose sense used to motivate the LAE experiments; it is a specific interval.

1.5 A hardware constraint that is easy to miss

Recovering a phase to a useful precision from a sampled waveform requires adequate sampling within each cycle. This sounds trivial and is not: an undersampled demodulation does not merely produce a noisy phase, it produces a wrong one, and the error does not announce itself as scatter around the true value. In our own analysis an estimate taken at one frame per cycle returned values of 164°, 266°, 194° and 254° against a true 68°. Any precision figure computed from such data is meaningless, and a design curve built from it will recommend operating points that cannot be realised.

We therefore derive and validate an explicit sampling requirement (≥ 5 frames per cycle) and show that the PAT-1M in oscillatory mode satisfies it with margin, while noting that the same instrument's static mode does not — a distinction that has caused confusion in the literature, because the two modes are reported in the same methods sections.

1.6 Contributions

1. A mass-conserving forward model of LAOD response that carries the micellar reservoir as an explicit immobile species with a linearised Aniansson–Wall exchange, so that total surfactant is conserved by construction rather than asserted. The model removes a structural defect of the two-compartment boundary-layer formulation in which the bulk reservoir never loses what the surface gains.

2. The detectable-versus-invertible distinction (§1.3), together with the demonstration that the harmonic ratio's sensitivity to τ_2 sits below its own noise while the phase lag carries the

information, and that in the LAOD regime the phase is dominated by higher harmonics rather than by the fundamental.

3. A measured concentration inversion window for $C_{12}E_8$, 1.2–3.0 × CMC with an interior optimum at 1.5 × CMC, and a demonstration that the optimum is invariant under a fourfold change in the assumed limiting elasticity E_0 — i.e. that the window is a property of the transport problem and not of the isotherm parameterisation.

4. An explicit sampling-density criterion (≥ 5 frames per cycle) with a demonstration that violating it biases rather than scatters the phase estimate.

5. Two inversion protocols with stated conditioning: a one-parameter fit (Protocol A), in which capacity is fixed by the independent CMC/aggregation-number closure and only τ_2 is estimated, and a multi-concentration regression (Protocol B), in which τ_2 and a capacity scale are fitted jointly. Protocol A gives τ_2 to ≈ 0.2 % on synthetic data and is the headline result. Protocol B is reported as a partial negative: it recovers τ_2 to 0.5–1.6 % but the τ_2–capacity correlation remains between 0.1 and 0.6, does not fall monotonically with the number of concentrations, and in one of three isotherm anchors gets worse when a second concentration is added. We state this because a protocol whose conditioning is misreported is worse than no protocol.

1.7 Scope and exclusions

We treat a single nonionic surfactant with a Langmuir equation of state. We do not treat lateral interactions, ionic double-layer effects, multicomponent mixtures, adsorption barriers, or curvature corrections to the pendant-drop geometry. The ionic surfactants are used only to demonstrate the predictive content of the timescale-to-bandwidth criterion of §6. We make no claim about the interpretation of the LAE second-harmonic anomaly; on the contrary, we show

that the model cannot reproduce it, and we treat that failure as evidence about LAE's compositional complexity rather than as a falsification of the micellar mechanism.

2. Coupled Reaction–Diffusion Model with Langmuir Interfacial Kinetics

2.1 Geometry, state variables, and what is conserved

We consider a planar interface at x = 0 bounding a semi-infinite surfactant solution. The bulk carries monomer at concentration c(x,t) and micellised surfactant at concentration M(x,t), both in mol m^{-3}. The interface carries surface excess Γ(t) in mol m^{-2}. The interface is driven by a prescribed area oscillation

$$A(t)/A_0 = 1 + \varepsilon \sin(\omega t), \tag{1}$$

with strain amplitude ε and angular frequency ω. The model returns γ(t), from which observables are extracted in §2.6.

The essential structural choice is that micelles are an explicit species rather than an absorbing boundary condition. This is what makes the mass balance a property of the discretisation rather than of a closure assumption.

2.2 Bulk transport and micellar exchange

$$\partial c/\partial t = D\, \partial^2 c/\partial x^2 - \partial M/\partial t, \tag{2}$$

$$\partial M/\partial t = -[\, (M - Cm\, c) - (Meq^0 - C_m\, c_0) \,] / \tau_2. \tag{3}$$

Equation (2) states that monomer is transported by Fickian diffusion and exchanged with the micellar pool. Equation (3) is the linearised Aniansson–Wall relaxation of the micellar population towards its local equilibrium value Meq(c) = Cm c, with relaxation time τ_2. Two points about the form of (3) deserve emphasis.

First, the exchange term is written in the conservative form. Because $\partial c/\partial t$ contains $-\partial M/\partial t$ with the same expression, the sum c + M is unaffected by exchange: whatever the micelles release the monomer gains, and vice versa. This is not cosmetic. The two-compartment formulation in common use prescribes a boundary-layer thickness $\delta = \sqrt{(\pi D/\omega)}$ and writes the subsurface flux as $(2D/\delta)(c_0 - c_s)$, which imposes a flux across a fixed slab and lets the bulk reservoir supply surfactant without ever being depleted. For a freshly created interface, or at the strain amplitudes considered here, the resulting mass error is first-order, not a small correction.

Second, the exchange is linearised about equilibrium with the offset made explicit in (3). Without the $(M_{eq}^0 - C_m c_0)$ term the equation relaxes towards equilibrium rather than about it, which produces a spurious steady drift superimposed on the oscillation.

The reservoir is characterised by exactly two parameters, and both are required:

• capacity $C_m = dM/dc$ (dimensionless) — how much surfactant the micelles can supply within a cycle. This is obtained from equilibrium data, not fitted: $C_m = (c_{tot} - CMC)\cdot \bar{n}/CMC$ with a micellar-volume correction, where $\bar{n}$ is the mean aggregation number. For $C_{12}E_8$, $\bar{n} = 95$.

• τ_2 — how fast they supply it.

A model containing capacity without τ_2, or τ_2 without capacity, cannot separate "strong reservoir" from "fast reservoir"; this is the central deficiency of the prototype formulation this work replaces.

Boundary conditions are no-flux for micelles at the interface (micellar diffusion is neglected; the micelles are much larger and slower than monomers and matter only at the highest frequencies) and Dirichlet $c = c_0$ at the far boundary, which is placed at $L = 8\sqrt{(D/\omega_{min})}$ so that the outer boundary never sees the oscillation.

2.3 Surface balance

$$d\Gamma/dt = D\,\partial c/\partial x|_0 - \Gamma\, d(\ln A)/dt. \quad (4)$$

The first term is adsorption from the subsurface; the second is the dilution of surface excess by area expansion. The interface is taken to be in local equilibrium with the subsurface concentration $c_s \equiv c(0^+,t)$, so that

$$\Gamma = \Gamma(c_s) \quad (5)$$

holds instantaneously. Equation (5) is algebraic, and how it is used numerically is the single most important implementation decision in the model.

Writing cs as the differential state — the "obvious" choice — is wrong. If cs is integrated alongside the bulk, the interface satisfies $\Gamma = \Gamma(cs)$ at every instant by construction and is therefore infinitely compressible: the model then reports an elasticity of order $1/E_0$ and no nonlinearity. The correct formulation carries Γ as the differential state and obtains cs algebraically by inverting (5). All results in this paper use the inverted form.

2.4 Equation of state

For a Langmuir layer, with $\Gamma\infty$ the saturation excess and K the adsorption equilibrium constant,

$$\theta = \Gamma/\Gamma\infty, \quad K\,c_s = \theta/(1-\theta), \quad (6)$$

$$\gamma = \gamma_0 + nRT\Gamma\infty \ln(1-\theta), \quad (7)$$

with n = 1 for a nonionic surfactant. Linearising (7) about the static point gives the limiting elasticity

$$E_0 = nRT\Gamma\infty \cdot \theta/(1-\theta). \quad (8)$$

Two features of (8) must be stated explicitly because they are frequent sources of error. E_0 diverges as $\theta \to 1$; the prefactor $nRT\Gamma\infty$ is not a ceiling on the elasticity but a scale. And,

conversely, an assumed E_0 pins the operating coverage: a typical measured E_0 of 20–40 mN/m for a single-chain surfactant corresponds to θ between roughly 0.6 and 0.85, not to the $\theta \approx 0.95$ that large fitted adsorption constants tend to imply. Parameter sets whose K places the monolayer in the saturated regime propagate an unphysical elasticity into every downstream number, and we recommend (8) be used as a routine consistency check on any reported (K, $\Gamma\infty$) pair.

Because this paper is restricted to $C_{12}E_8$, interactions are set to zero and (6)–(7) are used throughout. The generalised (Frumkin) boundary used in Paper 2 replaces $\ln(1-\theta)$ by $\ln(1-\theta) + a\theta^2$ in (7), which changes (6) to the implicit isotherm

$$K\, c_s = [\theta/(1 - \theta)] \cdot \exp(-2a\theta). \tag{9}$$

We record (9) here only to fix the sign convention: the interaction term enters the isotherm with a negative exponent and enters the equilibrium condition with the opposite sign, and a residual with the sign reversed will converge to a spurious root. For $a > 2$ the isotherm is multivalued and the branch choice becomes a physical assumption; this bounds the model and is deferred to Paper 2.

2.5 The natural dimensionless group

The problem has two timescales: the micellar relaxation τ_2 and the experimental period $2\pi/\omega$. Their ratio is the controlling group,

$$\Omega \equiv \omega\tau_2, \tag{10}$$

and the transport problem is additionally characterised by the diffusion time implicit in the inverse-isotherm slope, $\omega D = D(\partial cs/\partial\Gamma)^2$. For $C_{12}E_8$ at 1.5 × CMC, $\tau D = 1/\omega D \approx 6$ ms against $\tau_2 = 4$ s: the diffusion time is shorter by a factor of 634 (about 2.8 decades), so on the timescale of micellar exchange the interface is quasi-equilibrium with respect to diffusion, and τ_2 is the

slow process that sets the phase lag. (We note that for the fast ionic surfactants of §6 the ratio grows to three to four decades.)

We note explicitly, because we tested and refuted it, that Ω alone does not collapse the response across surfactants: at matched Ω the fundamental amplitude differs by a factor of 2.4 between $C_{12}E_8$ and CTAB. No single-group scaling law is claimed.

2.6 Observables

Fourier decomposition of the steady-state response over an integer number of cycles,

$$\gamma(t) = \bar{\gamma} + \Sigma k \, [\, ak \cos(k\omega t) + b_k \sin(k\omega t) \,], \tag{11}$$

defines the harmonics $Ik = \sqrt{(ak^2 + b_k^2)}$. We report:

• I_1, the fundamental amplitude (the linear response, $\approx \varepsilon|E|$);

• I_2, the second harmonic, the measure of nonlinearity, and the detectability statistic;

• the harmonic ratio I_2/I_1, the quantity most often reported in the literature and, for the purposes of inversion, effectively uninformative (§1.3);

• the phase lag φ, defined as the lag of γ behind A:

$$\varphi = \text{phase}(A) - \text{phase}(\gamma), \quad \text{positive when } \gamma \text{ peaks after A.} \tag{12}$$

Sign convention (resolved). We define the loss angle δ as the lag of γ behind A, positive when γ peaks after A. In the linear limit a diffusion-controlled soluble monolayer must give $\delta \to 0$ as $\omega \to \infty$ (insoluble limit: the surface excess cannot follow the area, so γ responds elastically and in phase) and $\delta \to 45°$ as $\omega \to 0$. Under the tabulated LVDT modulus $\varepsilon(\omega) = \varepsilon_0(1+\xi+i\xi)/(1+2\xi+2\xi^2)$ with $\xi = \sqrt{(\nu_D/2\nu)}$, the loss angle is $\delta = \arctan[\xi/(1+\xi)]$, which is bounded by 45° as $\xi \to \infty$. The model's harmonic definition reproduces this expression identically.

We verified the correspondence rather than assuming it. With Cm = 0 and νD computed self-consistently from the isotherm, $\nu D = D(\partial cs/\partial\Gamma)^2/2\pi$ = 0.500 Hz for $C_{12}E_8$ at 0.5 × CMC, model and LVDT agree to 0.48° at worst over 0.002–20 Hz (e.g. 18.78° vs 18.43° at 1 Hz). An earlier apparent 32°-vs-18.8° discrepancy was an artefact of comparing against a ν_D imported from an unrelated system; it is not a convention conflict, and no offset correction is applied anywhere in this paper.

This 45° ceiling is the central diagnostic of the paper. As §4 shows, a solution containing micelles routinely produces δ above 45° — up to 80° for $C_{12}E_8$ — which is impossible for any diffusion-only monolayer at any frequency. The excess above 45° is therefore a categorical signature of a micellar reservoir, and its magnitude is what carries τ_2.

2.7 Numerical method and what has been verified

Equations (2)–(4) are integrated as a differential-algebraic system in the state vector [Γ, $c_1 \ldots c\{N-1\}$, $M_1 \ldots M\{N-1\}$], with c_s recovered algebraically from (5). Space is discretised by a finite-volume scheme on a geometrically stretched grid, N = 96 cells, with cell widths expanding as $(\exp(\sigma\xi)-1)/(\exp(\sigma)-1)$ for σ = 4, so that resolution is concentrated at the interface where the gradient is steepest. Time integration uses an implicit BDF scheme at rtol = 10^{-7}, atol = 10^{-11} with a step ceiling of T/160. The last four of N ≥ 12 cycles are retained as steady state, and harmonics are extracted by least squares over an integer number of cycles.

Three verifications are performed and reported in the SI:

1. Mass conservation. The relative drift of ∫c dV + ∫M dV + Γ over a run is computed, not assumed. It is 2.2 × 10^{-3} at the operating point; the residual is attributable to exchange across the semi-infinite Dirichlet boundary, which the domain can legitimately exchange with, and is monitored for that reason.

2. Grid and cycle convergence against N ∈ {64, 96, 144} and N_cycles ∈ {12, 18, 24}.

3. The no-micelle limit. Setting C_m = 0 must recover diffusion-limited behaviour; the phase lag must then be τ_2-independent, which provides a null test for the inversion machinery.

2.8 Deliberate omissions

The model neglects: micellar diffusion; any adsorption or desorption barrier (the interface is at local equilibrium); lateral interactions (a = 0 throughout this paper); pendant-drop curvature and capillary depletion; and any heterogeneity in the adsorbed layer. Each omission is either immaterial in the regime studied or is bounded by an explicit test, and each is a candidate mechanism for the discrepancy between model and the LAE data discussed in §5. The model is not offered as a general description of surfactant dynamics; it is offered as the minimal structure that contains τ_2 and capacity as separately identifiable quantities.

3. The Concentration Inversion Window

3.1 Why a window must exist, and why it is not where one expects

Above the CMC the micellar reservoir grows with total concentration. The intuitive expectation is monotone improvement: more reservoir, more buffering, stronger micellar signature. Both edges of the observed window contradict that expectation, and the reason is that capacity enters the problem twice with opposite effects.

• It buffers the subsurface. A larger Cm holds cs closer to equilibrium during the oscillation cycle, which is the micellar effect and should grow with concentration.

• It suppresses the nonlinearity that generates I_2. The second harmonic arises from the curvature of the isotherm traversed during the cycle. By holding the interface near equilibrium, a large reservoir makes the trajectory quasi-linear, and I_2 collapses faster than the phase effect grows.

The first effect sets the lower edge, the second sets the upper edge. Consequently the method lives in a bounded interval, and — the point that matters for replication — the informative region is not "near the CMC" in the loose sense in which that phrase is used to motivate large-amplitude experiments on micellar systems.

3.2 The window, measured

We fix $\tau_2 = 4.0$ s (the $C_{12}E_8$ benchmark value), drive at $\varepsilon = 0.20$, and sweep concentration at $\omega = 6.28$ rad/s (1.0 Hz, the top of the PAT-1M oscillatory band). Two criteria are applied simultaneously, as demanded by §1.3: I_2 above the 0.05 noise floor (detectability), and a change in loss angle of at least 3° per factor-of-four in τ_2 (invertibility).

Table 1. Summary of second-harmonic response and loss-angle sensitivity across the critical micelle concentration for $C_{12}E_8$.

c/CMC	θ	I_2	δ (°)	Δδ per 4× τ_2	detectable	verdict
1.0	0.909	0.129	32.4	0.0	yes	reservoir empty — τ_2-invisible
1.2	0.923	0.052	50.4	17.4	yes	usable
1.5	0.937	0.183	67.9	17.4	yes	optimum
2.0	0.952	0.152	78.7	10.4	yes	usable
3.0	0.968	0.080	84.3	4.4	yes	marginal upper edge
5.0	0.980	0.035	86.8	1.2	no	below noise floor
10.0	0.990	0.012	88.0	0.4	no	saturated

The window is 1.2–3.0 × CMC with an interior optimum at 1.5 × CMC. It is not a plateau: Δδ varies by a factor of 4 across the usable interval, so operating-point choice changes the achievable precision by the same factor. The lower edge is sharp — at exactly 1.0 × CMC, $\Delta\delta = 0.00°$ while I_2 remains large (0.129), a textbook instance of the detectability/invertibility distinction of §4: the nonlinearity is strongest where the reservoir is absent.

The upper edge is a genuine cliff, not a gentle roll-off: between 3.0 and 5.0 × CMC the second harmonic falls from 0.080 to 0.035, crossing the noise floor, and Δδ falls from 4.4° to 1.2°.

3.3 The window is a property of the transport problem, not of the isotherm

The results of §3.2 are computed at a specific isotherm parameterisation (K = 140 m³/mol, which at 1.5 × CMC gives θ = 0.937 and E_0 = 103.5 mN/m). Since E_0 is itself uncertain by a factor of several and §2.4 warns that an assumed E_0 pins the operating coverage, this is a potential confound. We therefore re-ran the entire sweep at four anchor elasticities spanning 25–103.5 mN/m, each corresponding to a different K and different coverage:

Table 2. Invariance of the concentration inversion window across a fourfold range in limiting surface elasticity.

Anchor E_0 (mN/m)	θ at 1.5×	K (m³/mol)	optimum	Δδ there (°)	usable range
25.0	0.783	33.8	1.5×	19.4	1.2–3.0×
45.3	0.867	61.3	1.5×	18.8	1.2–3.0×
73.9	0.914	100.0	1.5×	18.0	1.5–3.0×
103.5	0.937	140.0	1.5×	17.4	1.2–3.0×

The optimum is 1.5 × CMC at every anchor, and the leverage varies by only 11% across a fourfold range in E_0. We regard this as the single most important robustness result in the paper: the window is set by the competition between reservoir buffering and isotherm curvature (§3.1), both of which are properties of the transport problem, and neither of which depends on the particular (K, Γ∞) pair used to place the operating point on the isotherm.

We state the corollary plainly, because it constrains how the result may be cited: the window edges are robust, the absolute numbers are not. A paper quoting "Δδ = 17.4° at 1.5 × CMC" must state its E_0 anchor. We therefore quote Δδ with its anchor throughout, and report the range 17.4–19.4° rather than a single figure.

3.4 What the window is not

• It is not a statement about where micellar exchange is physically strongest. Exchange is strongest as $C_m \to \infty$. The window is where exchange is most measurable, which is a different and much narrower question.

• It is not transferable between surfactants without recomputation. §6 shows that the equivalent window for CTAB sits at frequencies the PAT-1M cannot reach; the shape of the argument transfers, the numbers do not.

• The upper edge depends on the noise floor, and therefore on instrument and averaging. With $\sigma_\gamma = 0.05$ mN/m we find the floor crossed between 3 and 5 × CMC. A tenfold improvement in noise would move that edge, and we have not computed where.

3.5 A cautionary note on the CMC itself

The window is expressed in units of CMC, so its location inherits any error in the CMC. For $C_{12}E_8$ we use 0.071 mM. Because the lower edge is sharp (§3.2), a 10% error in the CMC displaces the entire window by 10% and could place the nominal "1.5 × CMC" operating point outside the usable interval at the low end. The CMC must be measured, not taken from a table, before the protocol is applied to a new system. This is an unglamorous point that we expect to be the most common cause of failure to reproduce the method.

4. Observables: Detectability versus Invertibility

4.1 The loss angle and the 45° ceiling

Section 2.6 establishes that δ is the lag of γ behind A, and that the tabulated LVDT modulus gives $\delta = \arctan[\xi/(1+\xi)]$ with $\xi = \sqrt{(\nu_D/2\nu)}$, so that any diffusion-only monolayer has $\delta < 45°$ at every frequency, approaching 45° as $\omega \to 0$ and 0° as $\omega \to \infty$.

This bound is the organising fact for the whole paper. It is instrument-independent, it requires no fit, and it does not depend on E_0. A solution containing micelles routinely exceeds it.

At 1.5 × CMC and ω = 6.28 rad/s we obtain δ = 67.9° where the same system with C_m = 0 gives 37.9°.

A measured δ above 45° is therefore categorical evidence of a micellar (or other) reservoir, and no choice of D, E_0 or ν_D can produce it. The magnitude of the excess is what carries τ_2.

4.2 The reservoir raises the loss angle — and this is the opposite of the intuitive expectation

We initially expected the reservoir to make the interface more elastic, damping δ downwards from its diffusion-limited value, on the reasoning that rapid monomer replenishment holds the subsurface concentration constant and lets the surface excess follow the area more closely. Direct computation shows the opposite, and the sign matters enough to state explicitly:

Table 3. Frequency dependence of the dilatational loss angle with and without the micellar exchange reservoir.

f (Hz)	δ, C_m = 0	δ, C_m = 47.5	shift
0.010	44.28	51.81	+7.5
0.050	43.40	69.62	+26.2
0.200	41.83	79.98	+38.2
0.500	39.98	76.86	+36.9
1.000	37.91	67.91	+30.0

The shift is positive and large — up to +38° — and is not monotone in frequency: at fixed τ_2 = 4 s it peaks near 0.2 Hz and decreases on both sides. Nor is it monotone in τ_2: at ω = 6.28 rad/s the shift is +14.4° at τ_2 = 0.05 s, rises to +38.8° at τ_2 = 0.5 s, and falls again to +8.7° at τ_2 = 20 s.

The maximum therefore sits at intermediate $\omega\tau_2$ of order a few — approximately 3 on the available scan, not at $\omega\tau_2$ = 1 as a single-relaxation-time argument would suggest. We note this without explaining it: it is a property of the coupled surface/bulk problem rather than of the exchange term alone, and we did not attempt to derive the optimum. What matters operationally

is that the sensitivity has a maximum in the τ_2 direction as well as in concentration, so a system whose τ_2 lies far from the band centre is harder to invert for a second, independent reason.

The physical reading is that the reservoir is not primarily an elastic element. It is a lagging source term: the micelles deliver surfactant late in the cycle, when the subsurface has been depleted, which puts the γ response further behind the area drive and adds dissipation rather than restoring elastic energy. Modelling it as an added elasticity gives the wrong sign for the effect.

This also resolves the ambiguity in the phrase "phase depression" that appears in the LAOD literature: the reservoir raises δ; what lowers δ is the approach to the insoluble limit at high frequency. The two effects act in the same frequency range and are easily confused.

4.3 Detectability: I_2

I_2 measures the departure from linearity and sets the noise-limited reach of the method. It is driven by isotherm curvature and, as §3.1 shows, is suppressed by the reservoir. Its value as a criterion is thus asymmetric: a small I_2 rules the measurement out, but a large I_2 does not rule it in.

4.4 Invertibility: what actually carries τ_2

Three candidate observables, evaluated at the 1.5 × CMC optimum, as τ_2 is varied over a factor of 16:

Table 4. Comparison of candidate rheological observables across a 16-fold variation in micellar relaxation time τ_2 at 1.5 × CMC.

τ_2 (s)	I_1	I_2	I_2/I_1	δ (°)
1.0	1.485	0.131	0.0882	78.12
2.0	2.057	0.168	0.0816	75.19
4.0	2.734	0.183	0.0669	67.92
8.0	3.299	0.157	0.0475	57.75

16.0	3.563	0.112	0.0315	48.78

I_2/I_1 is not blind, but it is not resolvable. It changes by 0.007–0.019 per doubling of τ_2. The noise floor on the ratio, propagated from a 0.1 mN/m instrument reading on a 2–4 mN/m fundamental, is 0.025–0.05. The τ_2-induced change is therefore smaller than the measurement uncertainty, and a single-point inversion from the ratio cannot work. Any published inversion based on I_2/I_1 alone should be regarded as under-determined.

δ carries the signal. It moves 7.3–10.2° per doubling of τ_2 in the region τ_2 = 2–8 s, against a phase uncertainty of 0.18° achievable at adequate sampling (§5). The ratio of sensitivity to uncertainty is 40–57, versus 0.2–0.8 for the harmonic ratio — a separation of roughly two orders of magnitude. That is the quantitative content of the claim that the phase lag is the operative observable.

δ is not log-linear in τ_2. The response is 2.9° for 1→2 s, 7.3° for 2→4 s and 10.2° for 4→8 s, and flattens again beyond 8 s. Quoting a single "degrees per e-fold" would be wrong in both directions; the local slope near the operating point is the correct figure.

4.5 The high-harmonic caveat

At ε = 0.20 the fundamental response is strongly nonlinear (I_1 = 1.5–3.6 mN/m), and the waveform carries substantial power above the second harmonic. Two consequences:

1. δ, extracted from the fundamental only, is a projection of a nonlinear waveform, not a loss angle in the linear-response sense. Its τ_2 sensitivity is real and monotone (Table 4), but it should not be interpreted as the phase of a linear transfer function. We report it as an operational observable whose calibration is the model, not as a material property.

2. The strong higher-harmonic content means the observation is genuinely in the large-amplitude regime, which is the point — but it also means the analysis cannot be validated against linear LVDT theory beyond the C_m = 0 limit of §2.6, where we obtained 0.48° agreement.

An experimenter wishing to use δ as a material quantity would reduce ε until the harmonics are negligible. We tested what that costs and did not find a penalty in δ: over ε = 0.03 –0.30 the loss angle is constant to 0.17° (67.97° at ε = 0.03, 67.92° at ε = 0.20). The reason is that in this system the nonlinearity is small in the fundamental ($I_2/I_1 \approx 0.07$) and the C_m ≠ 0 response is already far from linear theory at every ε we can measure; the curved isotherm trajectory that produces I_2 is not what produces the phase lag.

The trade-off that does bind is therefore not δ versus linearity, and we correct an earlier draft that claimed it was. The binding constraint is signal-to-noise on I_2 for the detectability criterion: I_2 scales as ε^2 while δ does not depend on ε, so ε is set by how far I_2 must be lifted above the noise floor, and δ is then estimated from a waveform that happens to be nonlinear. This is a more favourable situation than we first supposed, and §5 quantifies it.

4.6 Summary of the two criteria

Table 5. Diagnostic comparison of detectability and invertibility criteria for micellar oscillatory dilatational rheology.

	Detectable	**Invertible**
Observable	I_2	δ
Noise-limited by	σ_γ on the fundamental	sampling density, then σ_γ
Behaviour vs. reservoir	decreases with C_m	increases with C_m
Behaviour at 1.0 × CMC	maximal ($I_2 = 0.129$)	zero ($\Delta\delta = 0.00°$)
Behaviour at 10 × CMC	below noise	saturated, τ_2-insensitive
Sensitivity vs. uncertainty	0.2–0.8	40–57

The two criteria peak at different concentrations: detectability at the CMC, invertibility at 1.5 × CMC. The operating point must be chosen for invertibility, which means deliberately

operating where I_2 is not maximal. This is the practical content of §1.3, and it is why a "maximise the nonlinear signal" design rule would select the one concentration at which the method cannot work.

5. Sampling Density and Experimental Protocol

5.1 The sampling requirement is a bias problem, not a noise problem

It is natural to treat temporal sampling as a signal-to-noise question and apply a Nyquist-style rule: two samples per cycle at minimum, more is better. For the present purpose that framing is dangerously wrong, because the quantity being extracted is a phase, and an under-sampled phase estimate is not merely imprecise — it is biased, and the bias does not average out over repeated measurements.

We quantified this directly. A waveform with a known phase lag of 67.9° was sampled and demodulated at varying densities, with no added measurement noise, the only error being the sampling itself:

Table 6. Demodulation accuracy and phase-lag recovery bias as a function of optical sampling density.

frames / cycle	**recovered δ**	**scatter**
1	164.3°, 266.2°, 194.1°, 254.1° (four sampling phases)	≈ 90°
2	—	43°
5	67.9°	0.33°
20	67.9°	0.18°

At one frame per cycle the returned value is not a noisy measurement of 68°; it is a number that bears no relation to it, and which varies by ±90° depending on where the single sample happens to fall. We report this because it affected our own analysis: an earlier design curve of ours, built on a 1 frame/s assumption, recommended an operating point at 1 frame per cycle with an apparent relative error of 4.5%. That figure was an artefact of favourable sample placement and is withdrawn.

The mechanism is aliasing of the higher harmonics together with the lost phase reference; the fundamental is not the only component present (§4.5), so a two-point demodulation is not merely under-determined, it is inconsistent with the waveform it is meant to describe.

5.2 The criterion

At least five frames per cycle. At five frames the phase is recovered to 0.33° and the estimate is unbiased; at twenty it improves only to 0.18°, so the returns saturate quickly once the catastrophic regime is escaped. We recommend 5 as the specification and 10–25 as the practical target, since the criterion must hold at the highest frequency in a sweep.

5.3 The PAT-1M satisfies this, and the confusion that says otherwise

The Sinterface PAT-1M in oscillatory mode acquires at 25–50 frames per second [8]. At 1.0 Hz — the top of its oscillatory band and the operating point of §3 — that is 25–50 frames per cycle, comfortably above the criterion. At 0.2 Hz it is 125–250.

A frequently cited figure of approximately one profile per second refers to the instrument's static drop-shape-tensiometry mode, in which a full profile is fitted to obtain a single equilibrium tension. The two modes are reported in the same methods paragraphs and the numbers are easily conflated; we note it because reading the 1 s figure as a sampling rate would imply that the PAT-1M violates the criterion by a factor of 25, and would wrongly rule out the only widely available instrument that satisfies it.

5.4 Strain amplitude: why ε = 0.20

The second harmonic scales as the square of the strain, which we verified rather than assumed. Over ε = 0.03–0.30 at 1.5 × CMC and ω = 6.28 rad/s, the local logarithmic slope of I_2 against ε is 2.00, 1.99, 1.95 and 1.84 at ε = 0.05, 0.10, 0.20 and 0.30 respectively — quadratic at

small amplitude with mild saturation at the top of the range. Going from Czakaj's $\varepsilon \approx 0.065$ to $\varepsilon = 0.20$ raises I_2 from 0.0198 to 0.1827 mN/m, a factor of 9.21 against the 9.5 predicted by pure ε^2.

That factor is what lifts the second harmonic clear of the 0.05 mN/m noise floor. At $\varepsilon = 0.065$ the $C_{12}E_8$ operating point gives $I_2 = 0.0198$, which is below the noise floor; the measurement would fail the detectability criterion outright even though the underlying physics is unchanged.

Crucially, δ itself is unaffected: over the whole $\varepsilon = 0.03$–0.30 range the loss angle moves by 0.17° (67.97° to 67.80°). The strain amplitude buys detectability and costs nothing in invertibility. An earlier draft of ours claimed the opposite — that operating at large amplitude traded linear-theory validity against τ_2 sensitivity — and that was wrong (§4.5, corrected). The loss angle here is an operational observable calibrated by the forward model, not a linear material property, and it is stable across a sixfold amplitude range.

5.5 The protocol

For a new nonionic surfactant of unknown τ_2:

1. Measure the CMC. Do not take it from a table. The window of §3 is expressed in units of CMC and its lower edge is sharp; a 10% CMC error displaces the whole window by 10%.

2. Determine $\bar{n}$ (aggregation number) independently, to fix the capacity closure $Cm = (c_{tot} - CMC)\cdot\bar{n}/CMC$. This is the only external input the inversion requires.

3. Set $c = 1.5 \times CMC$. This is the interior optimum of §3.2, invariant across a fourfold range in E_0.

4. Set $\varepsilon = 0.20$, subject to §5.6.

5. Sweep frequency across the instrument band, verifying that frames/cycle ≥ 5 at every point. Retain ≥ 12 cycles and analyse the last four.

6. Estimate δ at each frequency by least-squares Fourier decomposition over an integer number of cycles, then invert τ_2 by Protocol A (§7) using the Δ-method or a joint fit.

5.6 Precision budget, and what it costs

At the operating point (1.5 × CMC, ω = 6.28 rad/s, σ_γ = 0.05 mN/m, 60 Monte-Carlo realisations):

Table 7. Precision budget and parameter recovery uncertainty from 60 Monte Carlo realizations at the operating point.

step	**value**
Monte-Carlo scatter of recovered δ	0.033°
Monte-Carlo scatter at ω = 3.0 (for comparison)	0.065°
$\Delta\delta/\Delta\ln\tau_2$ at 6.28 rad/s	13.3° per e-fold
relSE by Δ-method	0.25%
relSE by joint 1-parameter fit	0.23%
relSE, 2-parameter fit (capacity also free)	3.0%, with corr(τ_2, C_m) = 0.997

We quote τ_2 to 0.2–0.3% under Protocol A, where capacity is fixed by step 2 and only τ_2 is estimated. The improvement from 0.065° at ω = 3.0 to 0.033° at ω = 6.28 is a factor of 2 for a factor of 2 in frequency, which is why the top of the instrument band is the operating point.

Under Protocol B, where capacity is also free, the estimate degrades to 3% with a correlation of 0.997 between the two parameters — i.e. essentially perfect degeneracy at a single concentration. No amount of averaging fixes that; only additional concentrations or an independent capacity do, and §7 reports how partially the former succeeds.

These are model-based figures with realistic amplitude noise. They exclude error in CMC, $\bar{n}$, and E_0, all of which enter the forward model and none of which is a measurement uncertainty in the inversion. Adding a 10% error in $\bar{n}$ would dominate the budget; we have not propagated it, and it should be treated as an open item rather than assumed small.

5.7 Honest hardware caveats

• ε = 0.20 is three times the amplitude of the published LAOD protocol it replaces. We have not verified drop stability at this amplitude on a 10 μL pendant drop with a 2 mm capillary; we have only the report that the instrument accommodates ε = 0.15–0.30 on standard capillaries [6,7]. This is a required bench validation and should be the first experiment performed. Nothing in the method survives if the drop detaches.

• 25–50 fps is quoted for oscillatory mode. Should a particular instrument/software combination acquire more slowly, §5.2 is the binding constraint and the method fails at the top of the band first.

• The 0.05 mN/m noise floor is our estimate, propagated from a 0.1 mN/m instrument precision on a 2–4 mN/m fundamental. A better calibration of σ_γ moves both the window's upper edge and the achievable precision.

6. The Timescale-to-Bandwidth Horizon

6.1 The design rule

Sections 3–5 establish that the method requires a reservoir that responds on the timescale of the drive. The relevant dimensionless group is $\Omega = \omega\tau_2$, and operability requires the drive frequency to sit in the band $\Omega \gtrsim 1$, with the sensitivity maximum near $\Omega \approx 3$ (observed, §4.2) and useful response out to $\Omega \approx 30$.

Because τ_2 differs by three orders of magnitude across common surfactants, this rule translates directly into an instrument requirement. For each system, the operable frequency band is

$$f \approx \Omega / (2\pi\,\tau_2), \quad \Omega \in [1, 30], \tag{13}$$

and the question of whether a given tensiometer can perform the inversion becomes a comparison of (13) with its bandwidth:

Table 8. Timescale-to-bandwidth horizon and instrument compatibility for nonionic and ionic surfactant systems.

surfactant	τ_2	operable band, eq. (13)	PAT-1M (0.005–1 Hz)	CPT (0.1–500 Hz)
$C_{12}E_8$	4.0 s	0.04–1.19 Hz	covers it (and 1.19 Hz exceeds the 1.0 Hz top of band by a factor of 1.2 — the optimum at 1.0 Hz is comfortably inside)	covers it
CTAB	65 ms	2.4–73 Hz	cannot reach	covers it
SDS	38 ms	4.2–126 Hz	cannot reach	covers most

This is the paper's predictive content outside its own system, and it is a statement about instruments, not about surfactants: the method is not limited to slow nonionics, it is limited to surfactants whose relaxation time the available bandwidth can straddle.

6.2 Why the PAT-1M fails for the fast ionics — the mechanism

It is not that CTAB lacks a micellar reservoir. It has one, with a capacity comparable to $C_{12}E_8$ at the same reduced concentration. The failure is that at 1 Hz the drive is far below the relaxation rate: $\Omega = 2\pi \times 1 \times 0.065 = 0.41$. The reservoir then tracks the area change almost quasi-statically, the interface stays near compositional equilibrium, and — exactly as in the over-buffered limit of §3.1 — the response goes flat.

The numbers show the suppression clearly. At 1.5 × CMC with E_0 anchored to 30 mN/m, CTAB's effective modulus I_1/ε at 1 Hz is 0.37 mN/m, against an insoluble-limit ceiling of 30. The interface is 98% buffered and has almost nothing left to say. Its second harmonic is 0.0057 mN/m, one-ninth of the noise floor. Sweep upward and the picture inverts:

Table 9. Frequency sweep of harmonic amplitudes and loss angle for CTAB showing dynamic unbuffering at high frequency.

f (Hz)	$\Omega = \omega\tau_2$	I_1/ε (mN/m)	I_2 (mN/m)	δ (°)
1.0	0.41	0.37	0.006	55.2
3.98	1.6	0.98	0.019	71.3

15.9	6.5	2.60	0.065	74.8
100	41	14.3	0.252	42.6

At $\Omega \approx 6.5$ the second harmonic reaches the noise floor, and the effective modulus has recovered by a factor of 7. CTAB would be measurable on a capillary-pressure tensiometer and is not measurable on a PAT-1M. That is the whole of the horizon result, and it is a one-line instrument specification rather than a limitation of the theory.

6.3 A historical consequence, offered as a prediction

There is a well-known asymmetry in the LAOD literature: micellar exchange is reported for slow nonionic systems far more often than for fast ionic ones. The usual explanation is experimental difficulty with ionic systems — foam, evaporation, surface charge.

Equation (13) offers a second and more specific explanation. A drop-shape tensiometer is intrinsically a low-frequency instrument (0.005–1 Hz), and a surfactant with τ_2 below ~0.1 s requires drive frequencies above its reach. Low-frequency drop-shape analysis is structurally incapable of observing micellar exchange in fast surfactants, independently of any chemical difficulty. We offer this as a testable prediction rather than a claim about any particular study: the prediction is that a fast ionic surfactant measured on a CPT at 10–100 Hz will show $\delta > 45°$, whereas the same system on a PAT-1M will not.

6.4 The 45° ceiling as an instrument-independent screen

The diagnostic of §4.1 gives §6 a second, cheaper application that requires no model at all. Because no diffusion-only monolayer can exceed $\delta = 45°$ at any frequency, a measured δ above 45° proves kinetic relaxation is present. The converse is weaker but useful: a measured δ below 45° does not prove the absence of a reservoir, only that the drive frequency was too low to see it.

This asymmetry is worth stating carefully, because it is the trap §6.2 describes. At 1 Hz, CTAB sits at $\delta = 55.2°$ — above the ceiling, so a reservoir would be detected — yet its I_2 is nine times below the noise floor, so τ_2 could not be inverted. The ceiling tells you a reservoir is there; it does not tell you the measurement carries enough information to quantify it. Detectability and invertibility again diverge, and this time across instruments rather than across concentrations.

We checked that the ceiling is not an artefact of our own model by computing the LVDT loss angle from Czakaj's own fitted (E_0, ν_D) pairs for the LAE system at 0.2–1.0 mM: every value falls below 33° at 0.01–1 Hz, as it must, since the LVDT form is diffusion-only by construction.

6.5 Limits of the horizon argument

• $\Omega \approx 3$ is observed, not derived. We did not solve for the optimum of the coupled surface/bulk problem, and we do not have a scaling law: at matched Ω, CTAB and $C_{12}E_8$ differ in fundamental amplitude by a factor of 2.4 (Section 6.5). Equation (13) is a design heuristic located by direct computation on three systems, not a collapse. Anyone applying it to a fourth surfactant should verify by running the forward model first.

• The CTAB and SDS parameters are anchored, not measured. E_0 = 30 and 22 mN/m were chosen from literature ranges, and the resulting coverages (θ = 0.72, 0.64) are lower than the saturated monolayers that measured surface excesses imply. We established analytically that the anchored isotherm is internally consistent, but it is not independently validated, and CTAB coverage is an open question requiring a measured Π–c isotherm.

• SDS does not clear the noise floor anywhere in the scanned range, reaching only 0.0048 mN/m at 10 Hz. That is a band-edge result — our scan stops at 100 Hz, where $\Omega \approx 24$, not a demonstration that SDS fails.

• The PAT-1M and CPT bandwidths are manufacturer specifications; neither instrument was operated in the course of this work.

7. Inversion Protocols and Concluding Remarks

7.1 The structural reason a single measurement is degenerate

Before presenting the protocols it is worth stating exactly why two of them are needed. At the 1.5 × CMC operating point the loss angle responds to the two reservoir parameters with almost identical lever arms and opposite signs:

$$\partial\delta/\partial \ln C_m = +14.1° \text{ per e-fold} \quad (14)$$

$$\partial\delta/\partial \ln \tau_2 = -13.3° \text{ per e-fold} \quad (15)$$

A larger reservoir raises δ; a faster reservoir lowers it. A single-frequency measurement therefore sees only a weighted difference, and the correlation between the two fitted parameters is 0.997 — not as a numerical accident, but as a direct consequence of (14)–(15). This is why the second-harmonic ratio is useless for the two-parameter problem regardless of how precisely it is measured, and why the problem is genuinely under-determined at one concentration.

The two protocols correspond to the two honest ways of breaking that symmetry: supply C_m independently, or supply additional concentrations.

7.2 Protocol A — one-parameter inversion (recommended)

Fix capacity from an independent equilibrium closure, $C_m = (c_{tot} - CMC)\cdot \bar{n}/CMC$, and estimate τ_2 alone from the frequency dependence of δ.

Inputs: a measured CMC and aggregation number $\bar{n}$. That is all. Design: $c = 1.5 \times CMC$, $\varepsilon = 0.20$, 5–25 frames per cycle, ≥ 12 cycles, analyse the last four, sweep the full instrument

band. Estimator: joint least squares over frequencies, or the Δ-method on the widest-lever-arm point.

Performance at $C_{12}E_8$, 1.5 × CMC, σ_γ = 0.05 mN/m: Monte-Carlo scatter of δ is 0.033° at ω = 6.28 rad/s; the lever arm is 13.3° per e-fold; the resulting relative standard error on τ_2 is 0.23% by joint fit and 0.25% by the Δ-method. τ_2 = 4.000 s is recovered exactly on synthetic data.

Conditioning: one parameter, well identified, no correlation to report. This is the protocol whose precision may be quoted.

7.3 Protocol B — multi-concentration regression (exploratory; documented partial negative)

Where $\bar{n}$ is unavailable, fit τ_2 and a capacity scale jointly across a concentration series in 1.2–2.5 × CMC.

Performance: τ_2 is recovered to 0.5–1.6% in every configuration we tested, across three isotherm anchors and for CTAB as well as $C_{12}E_8$. The correlation between τ_2 and C_m does not fall below 0.1, does not decrease monotonically with the number of concentrations, and in one of three anchors gets worse when a second concentration is added (0.60 → 0.40, with a sign change).

The correlation is also anchor-dependent rather than a fixed property of the protocol: the same concentration series gave a clean monotone improvement (−0.61 → −0.37 → −0.24) at one isotherm anchor and a non-monotone sequence at another. We therefore state the result as multi-concentration data partially breaks the degeneracy, by an amount that depends on where the concentrations sit relative to the window edges and which cannot be predicted from the number of points alone.

Practical guidance: four to five points spanning 1.2–2.5 × CMC is a reasonable design, but a two-point fit should be regarded as under-determined and should not be reported as a joint determination of τ_2 and C_m.

7.4 Systematic error dominates the precision budget

We emphasise what the headline figure does not include. Because δ responds to Cm at +14.1°/e-fold (14) and Protocol A takes Cm from $\bar{n}$, any error in $\bar{n}$ propagates directly:

Table 10. Systematic propagation of aggregation number uncertainty into micellar concentration and relaxation time bias.

$\bar{n}$ error	C_m error	induced δ bias	τ_2 bias
5%	5%	0.69°	5.2%
10%	10%	1.34°	10.1%
20%	20%	2.56°	19.3%

A 10% error in $\bar{n}$ produces a 10% bias in τ_2 — forty-four times the 0.23% statistical uncertainty. No amount of signal averaging reduces it. The precision of the method is therefore bounded by the accuracy of an equilibrium measurement made by an entirely different technique, and the appropriate headline for this work is not "τ_2 to 0.2%" but "τ_2 to 0.2% statistical, 1:1 systematic in $\bar{n}$".

We regard this as the single most important caveat in the paper, and it is the reason Protocol A is described as requiring a measured $\bar{n}$ rather than a tabulated one. The same argument applies to the CMC, which sets both the concentration axis and the capacity closure (§3.5).

7.5 What the method buys, and what it costs

Buys: a quantitative, model-calibrated route to τ_2 from an instrument already present in most interfacial laboratories, without a frequency sweep beyond 1 Hz and without an

independent micellar-kinetics measurement. The 45° ceiling is a free, instrument-independent presence test requiring no fitting at all.

Costs: a strain amplitude three times larger than the standard LAOD protocol ($\varepsilon = 0.20$ against 0.065), which requires bench validation of drop stability that we have not performed; a measured $\bar{n}$ and CMC; adequate sampling, which must be verified rather than assumed; and operation at a concentration chosen for invertibility rather than for signal strength.

7.6 Conclusions

1. The loss angle, not the harmonic ratio, is the operative observable. The ratio's sensitivity to τ_2 lies below its own measurement uncertainty; the loss angle's exceeds it by two orders of magnitude. Inversions based on harmonic ratios alone should be regarded as under-determined.

2. A 45° ceiling makes the micellar reservoir categorically detectable. Any $\delta > 45°$ proves kinetic relaxation is present, independent of D, E_0 or any fit. The reservoir raises δ rather than depressing it, acting as a lagging source term rather than an elastic element.

3. Detectability and invertibility peak at different concentrations, and the operating point must be chosen for the latter. The inversion window for $C_{12}E_8$ is 1.2–3.0 × CMC, optimum 1.5 × CMC, and is invariant across a fourfold range in E_0. At exactly 1.0 × CMC, I_2 is maximal and the τ_2 sensitivity is zero.

4. Sampling density is a bias constraint, not a noise constraint. Below five frames per cycle the phase is wrong, not merely imprecise. The PAT-1M at 25–50 fps satisfies this with margin; its static mode, sometimes quoted at one profile per second, does not.

5. The method converts into an instrument-selection criterion. Requiring $\omega\tau_2 \gtrsim 1$ means the 0.005–1 Hz band covers $C_{12}E_8$ but not CTAB or SDS, which need a CPT above ~16 Hz. This

is offered as the explanation for the historical absence of micellar-exchange reports from low-frequency drop-shape work on fast ionics.

6. Precision is statistical; accuracy is systematic. τ_2 is recovered to 0.2–0.3% for a known $\bar{n}$, but $\bar{n}$ errors propagate one-for-one and dominate by a factor of 44.

7.7 What this work does not establish

• It does not validate the model against a measured $C_{12}E_8$ LAOD dataset. Every quantitative figure is a forward-model result with injected realistic noise; the $\tau_2 = 4$ s benchmark is a literature value, not one we inverted from data. The single most valuable next experiment is a measured $C_{12}E_8$ frequency sweep at 1.5 × CMC and $\varepsilon = 0.20$ to test the predicted $\delta(\omega)$ curve directly.

• It does not explain the LAE second-harmonic anomaly. Our model cannot reproduce it (§3.4), and we treat that as evidence about LAE's compositional complexity rather than as a falsification of the micellar mechanism. No claim is made about compositional reorganisation.

• It does not establish the CTAB or SDS parameters. Those coverages are anchored to literature E_0 values, not measured, and the CTAB monolayer capacity is in tension with measured surface excesses (Section 6.5).

• It does not derive the $\omega\tau_2 \approx 3$ sensitivity maximum. That is an observation on three systems, and no scaling collapse exists — at matched $\omega\tau_2$, CTAB and $C_{12}E_8$ differ in amplitude by 2.4×.

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